Evaluating the Impact of Tsunamis on Vegetation and Its Recovery: The case of Sendai Bay (Japan)

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Abstract

The 2011 tsunami, triggered by a magnitude 9.0 earthquake off the coast of Tōhoku, caused significant devastation to coastal ecosystems and agricultural areas in Japan. This study evaluates the impact on vegetation and its recovery trajectory using NDVI data derived from Landsat 7 satellite imagery. We analyze vegetation dynamics across various flood zones from 2010 to 2023, isolating changes attributed to the tsunami and identifying factors influencing recovery.

1 Introduction

On March 11, 2011, a devastating tsunami caused by a subduction zone earthquake (McGill, 2013) struck Japan's Pacific coast. With waves reaching heights of up to 40 meters, the tsunami caused catastrophic damage, including the Fukushima Daiichi nuclear disaster. This work focuses on the impact on vegetation. Using remote sensing data, the aim is to quantify vegetation loss caused by the tsunami, analyze long-term recovery trends, and assess some reasons of these trends.

2 Methods

2.1 Data

Digital Elevation Model (DEM) data was obtained from the SRTM (USGS/SRTMGL1_003), while vegetation dynamics were analyzed using the Normalized Difference Vegetation Index (NDVI), calculated from Landsat 7 Bands B3 (Red) and B4 (Near-Infrared). The study focuses on two key periods: 2010-2011 for immediate impact analysis and 2011-2023 for long-term recovery trends.

Landsat imagery was used because Sentinel data is not available for the studied period. Although Sentinel would have been preferable due to its higher spatial resolution, Landsat provides the highest resolution data available for this timeframe. Additionally, the relatively high revisit frequency of the Landsat satellite is useful for analyzing temporal trends.

To address the issues caused by Landsat's black striping artifacts, corrections were performed using `fillGaps()` in Google Earth Engine

2.2 Image Segmentation

Creating an accurate mask of the flooded area posed challenges due to the unavailability of high-resolution data for the precise period just before and after the tsunami. As a result, it was difficult to delineate the exact extent of the inundation zone using change detection in NDWI. However, altitude serves as a relatively good proxy for flood vulnerability (Defossez et al., 2018) to tsunamis in coastal regions. While not entirely precise, altitude provides a representative approximation of inundation zones.

To quantify the tsunami's impact, we segmented the study area into five elevation categories based on the Digital Elevation Model (DEM) data:

- $0 - 5$ meters,
- $5 - 10$ meters,
- $10 - 20$ meters,
- $20 - 30$ meters,
- $30 - 40$ meters.

These categories were chosen to capture the gradient of vulnerability to flooding, with lower altitudes representing areas most likely to have been inundated. By analyzing the vegetation response within each elevation band, this segmentation allows us to evaluate the tsunami's impact on vegetation and subsequent recovery relative to flood vulnerability. This method, even simplified, presents insights into vegetation dynamics in the absence of detailed inundation maps.

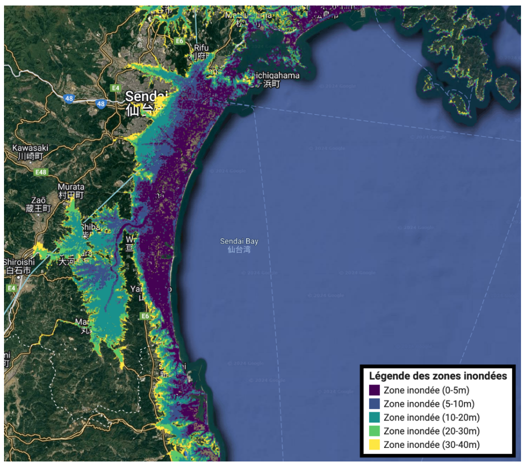

Figure 1: Elevation zones

2.3 NDVI and Differentiated NDVI

Normalized Difference Vegetation Index (NDVI): The NDVI is a remote sensing index that measures vegetation health and density. It is calculated as the normalized difference between the Near-Infrared (NIR) and Red bands of satellite imagery:

$$\text{NDVI} = \frac{\text{NIR} - \text{Red}}{\text{NIR} + \text{Red}}$$

Values range between -1 and 1, where higher values indicate dense, healthy vegetation, while lower values correspond to bare soil, water, or damaged vegetation.

In this work, the NDVI serves as an indicator to evaluate the health and density of vegetation across different flood zones and time periods. Landsat 7 imagery was used to compute annual mean NDVI values from 2010 to 2023, capturing trends in vegetation dynamics following the tsunami event.

Differentiated NDVI (ΔNDVI): The ΔNDVI represents the change in vegetation health between two time periods. It is calculated by subtracting the NDVI of one year from another:

$$\Delta\text{NDVI} = \text{NDVI}_{\text{after}} - \text{NDVI}_{\text{before}}$$

Positive values indicate vegetation gain, while negative values reflect vegetation loss.

the ΔNDVI was computed for two critical periods:

- **2010-2011:** To quantify vegetation loss immediately following the tsunami, highlighting areas most affected by flooding and soil salinization.
- **2011-2023:** To assess recovery dynamics and the effectiveness of restoration efforts over the long term.

The combination of NDVI and ΔNDVI allows for a nuanced analysis of the tsunami's impact on vegetation and its subsequent recovery. These mesures provide indicators for monitoring ecosystem health in tsunami-affected regions.

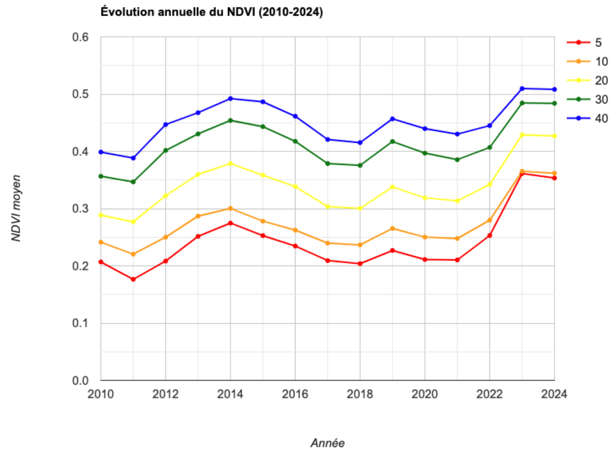

Figure 2: Mean NDVI for each area and each year

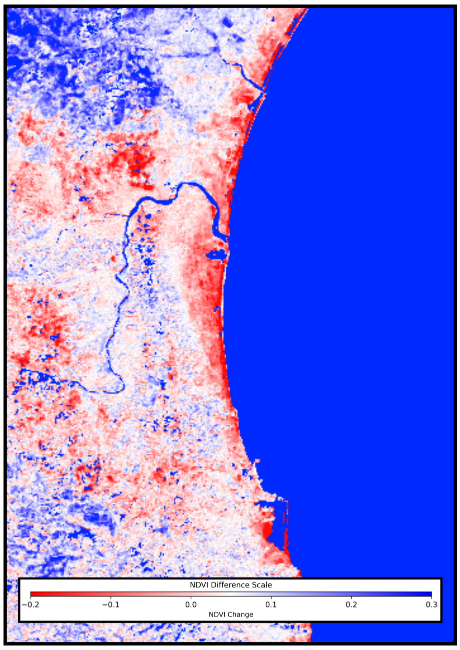

Figure 3: NDVI Difference before-after tsunami event (with mask on permanent water area).

3 Results

3.1 Immediate Vegetation Loss (2010-2011)

The ΔNDVI analysis reveals significant vegetation loss in coastal zones, particularly in areas between 0-5m and 5-10m elevation. The bar chart (Fig.4) shows a marked decrease of up to -0.12 NDVI in the most affected zones. Coastal regions experienced significant vegetation loss. In contrast, higher zones above 20 meters exhibited less change, highlighting the localized impact of the tsunami. The tsunami's impact on vegetation, as measured by ΔNDVI, is evident across all studied elevation zones. However, the magnitude of this effect diminishes with increasing altitude. Coastal areas at lower elevations (0–5m) exhibit the most significant changes in NDVI, while higher elevation zones (20–40m) show relatively less impact. This trend highlights the expected correlation between flood vulnerability and vegetation disturbance (Fig. 2, 3, 4).

We observe (Fig.2) that the slopes between consecutive years are generally steeper for lower elevation zones compared to higher ones. This suggests that vegetation in lower elevation zones is more susceptible to disturbances, experiencing both sharper declines and faster recoveries. This finding reinforces the idea that lower zones are more vulnerable to environmental perturbations, such as the tsunami, while higher zones exhibit better stability over time.

3.2 Vegetation Recovery (2011-2023)

A significant recovery in NDVI is observed between 2011 and 2023, particularly in lower elevation zones. The 0-5m zone showed an average increase of $+0.3$ NDVI, while higher zones between 20-40m remained more stable, suggesting natural resilience and minimal tsunami impact. The bar chart (Fig. 4) illustrates this gradual vegetation recovery.

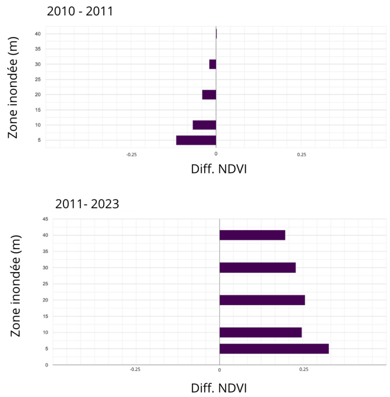

Figure 4: NDVI Difference in each altitude area.

It is also remarkable to observe (Fig.2) that vegetation in all zones reaches a higher NDVI in 2023 than before the disaster. This difference is particularly pronounced in the lower altitude zones. This is confirmed with Diff. NDVI (Fig. 4).

4 Discussion

The patterns of NDVI loss and recovery highlight the 2011 tsunami's impact on Japan's coastal vegetation and the factors influencing post-event recovery. This section examines these trends, supported by literature, outlines limitations and future possible improvements.

4.1 Drivers of NDVI Loss and Recovery

The significant NDVI reduction in low-altitude zones (0–5m) post-tsunami is primarily due to soil salinization and physical destruction. The tsunami deposited saline water and sediments onto coastal lands, rendering soils temporarily unsuitable for plant growth (Chagué-Goff et al., 2013). This salinization delayed vegetation recovery, particularly in coastal zones (Hara et al., 2016). Additionally, the mechanical force of the tsunami waves uprooted vegetation, stripped topsoil, and disrupted ecosystems, with effects most pronounced near the coastline (Hara et al., 2016). Higher altitudes experienced less flooding, resulting in less disturbance and a gradient of NDVI loss across elevation bands.

Vegetation recovery, especially in lower zones, reflects natural regeneration and human intervention. Grasses and pioneer species recolonized affected areas, demonstrating resilience (Oka et Hirabuki, 2014). Restoration efforts, including reforestation, crop reintroduction, and soil restoration, significantly contributed to recovery. Actions to reduce soil salinity and replant pines (GlobalGiving, 2024) after 2015 had substantial impacts (Hara et al., 2016). Seasonal variability and typhoons further influenced NDVI changes, adding stress to recovering areas and complicating NDVI interpretation.

4.2 Interpreting Elevated NDVI in 2023

In 2023, NDVI levels surpassed pre-tsunami measurements across all zones, particularly at low altitudes. Improved land use practices likely contributed, with more resilient cropping systems enhancing NDVI values. Reforestation increased canopy density, further boosting NDVI. Changes in land cover, such as vegetation replacing bare soil, also played a role. These outcomes reflect successful restoration efforts; however, land use changes may have introduced biases (ENS de Lyon, 2021).

4.3 Limitations and Possible Improvements

This study has limitations. Landsat 7's 30-meter resolution may miss fine-scale changes, particularly in mixed land cover areas. Higher-resolution imagery could improve spatial accuracy. External factors such as typhoons, weather variations and human activities complicate isolating tsunami impacts. While NDVI is a good proxy for vegetation health, it cannot distinguish between natural and anthropogenic changes without additional data.

Using altitude-based zones as a flood area is a simplification. More precise flood masks from radar data, high resolution images or hydrological surveys could refine the analysis. Landsat 7's 16-day revisit period also limits temporal coverage, potentially missing short-term changes for understanding immediate impacts.

Machine learning techniques can address these challenges. Anomaly detection and causal models can isolate tsunami effects from other influences. Unsupervised learning can identify recovery patterns, while supervised models can predict recovery based on environmental and human factors. These tools would enhance precision and allow a better understanding of ecosystem resilience to natural disasters.

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